

ELECTRIC VEHICLE DATA ANALYSIS

Analyse the overall landscape of electric vehicles to evaluate market's size and growth by using total vehicles, avg electric range and by comparing Battery Electric Vehicle (BEV) and Plug-in Hybrid Electric Vehicle (PHEV) counts and percentages.

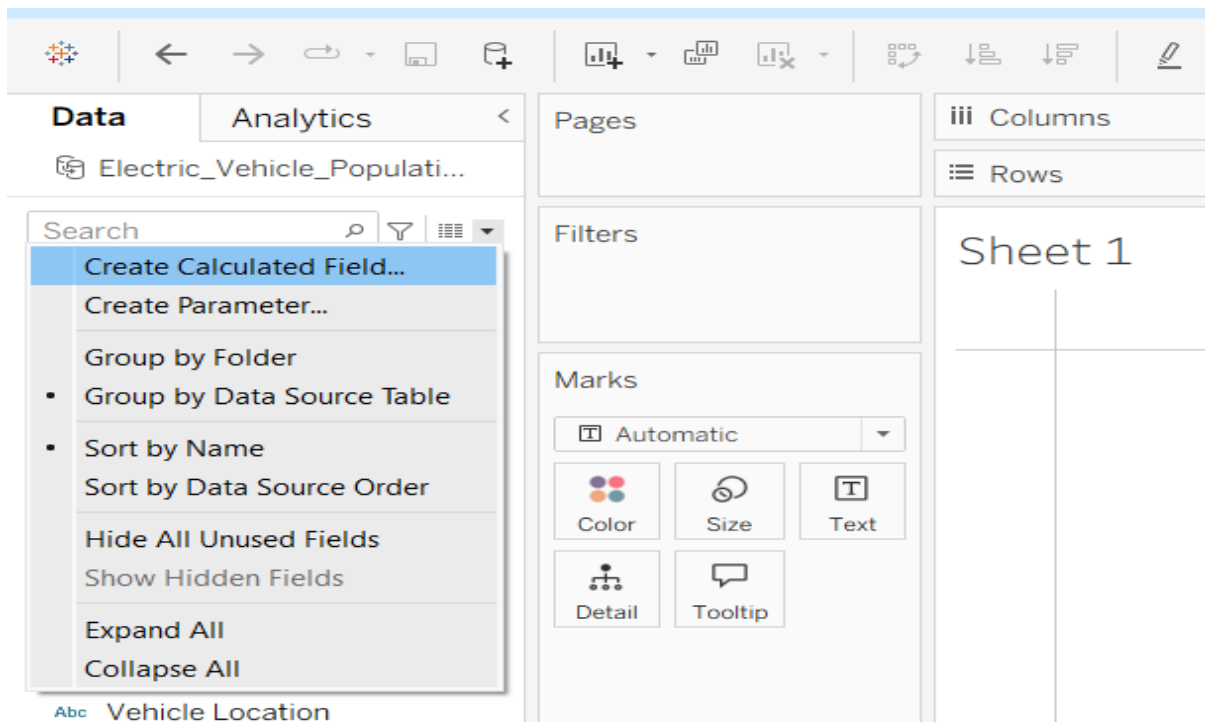
Step 1: Connect to Data

- Open Tableau then Click on **Connect** on the left sidebar.
- Choose data source by selecting **text file** and load dataset into Tableau.

Before going to the sheets creation, we have to create some **calculated fields** (new data fields) by applying custom formulas.

Step 2: Create Calculated data fields

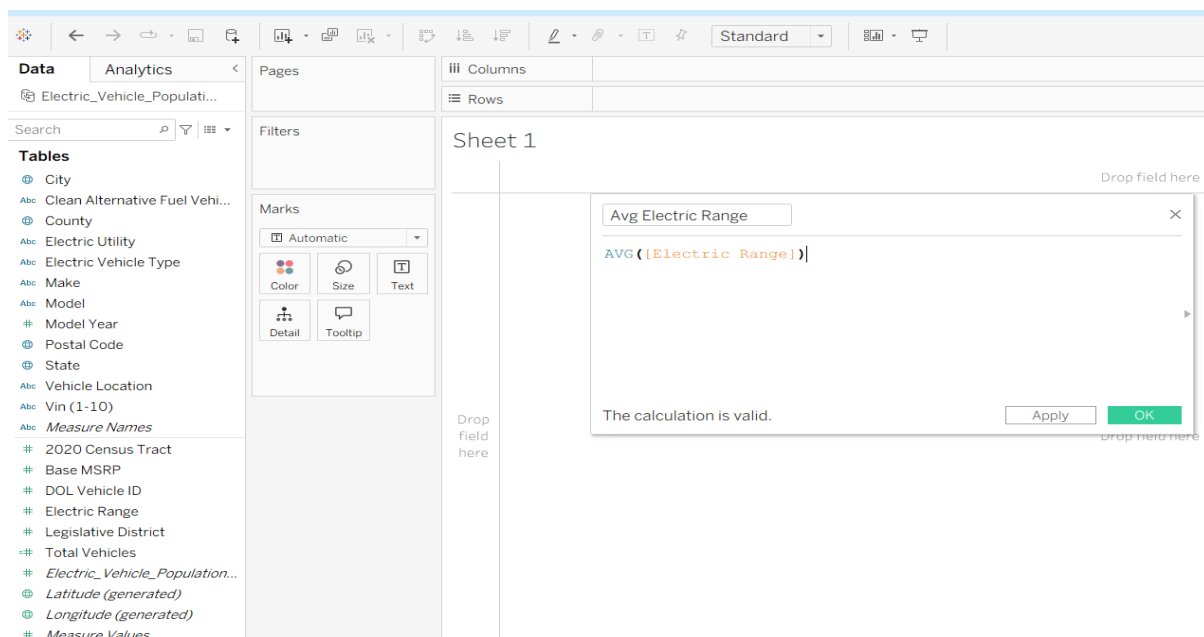
- Click on the small dot like **down arrow** on left side search bar, above the Tables.
- Select **Create calculated fields**.



- Name the field as **Total Vehicles**. Then add **custom formula** shown below.
- After the calculation field is valid then click on **Ok**.
- Total Vehicles data field is added to the Tables bar.

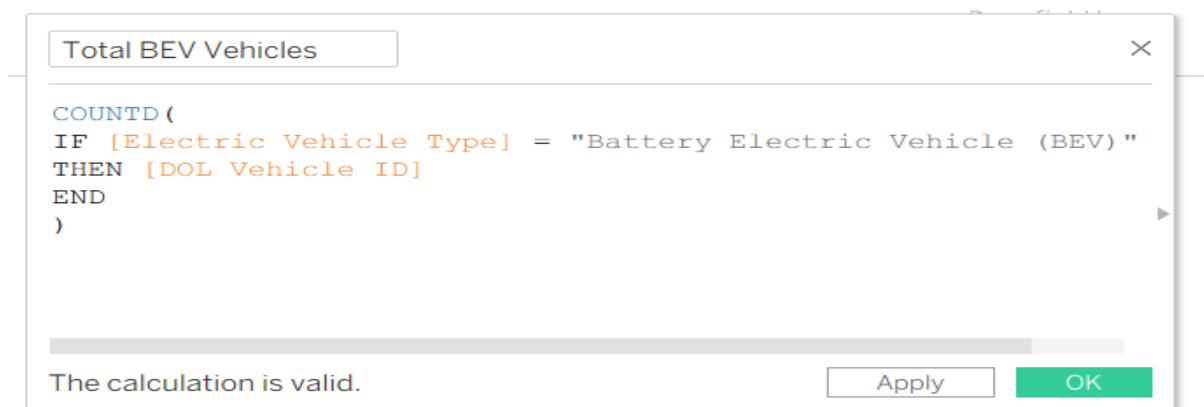


Similarly create calculated fields for **Avg Electric Range** By applying below shown custom formulas.



In above picture we can see that previously added **Total Vehicles** data field is added on Tables bar.

For creating BEV data field we have to create **percentage of BEV Vehicles** with the **Total BEV Vehicles**. Because we have to show that how many vehicles are operating on **Battery**.



% of BEV Vehicles

[Total BEV Vehicles] / [Total Vehicles]

The calculation is valid.

Apply OK

Now we have to create **percentage of PHEV Vehicles** and **Total PHEV Vehicles** data fields.

Total PHEV Vehicles

```
COUNTD(  
IF [Electric Vehicle Type] = "Plug-in Hybrid Electric Vehicle (PHEV)"  
THEN [DOL Vehicle ID]  
END  
)
```

The calculation is valid.

Apply OK

% of PHEV Vehicles

[Total PHEV Vehicles] / [Total Vehicles]

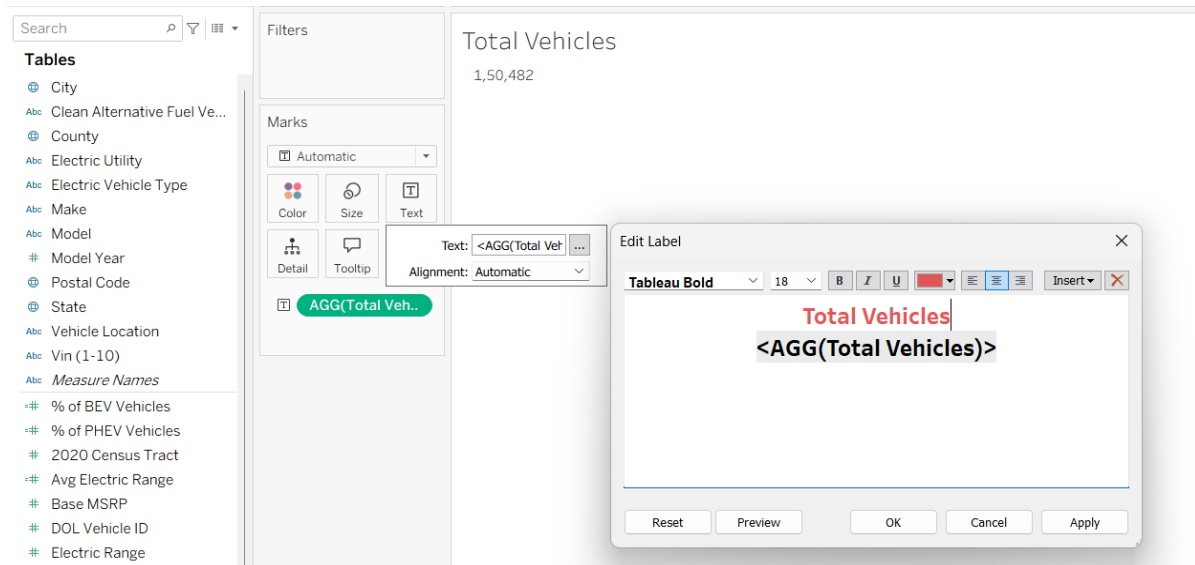
The calculation is valid.

Apply OK

Step 3: Create Required Sheets

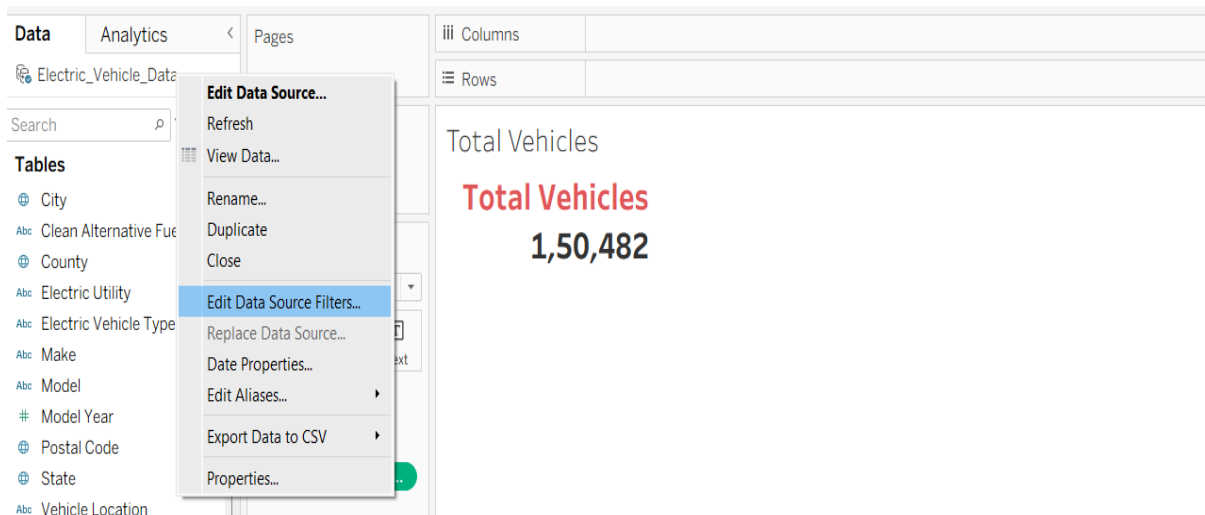
Sheet 1:

- Name Sheet1 as **Total Vehicles**. Drag Total vehicles data field into **Text** on Marks bar.
- Click on the Text Then Select **three dots (...)**, **Edit Label** tab will be opened.
- In edit label we can add Sheet Name for easy Recognition Then Click on **Ok**.

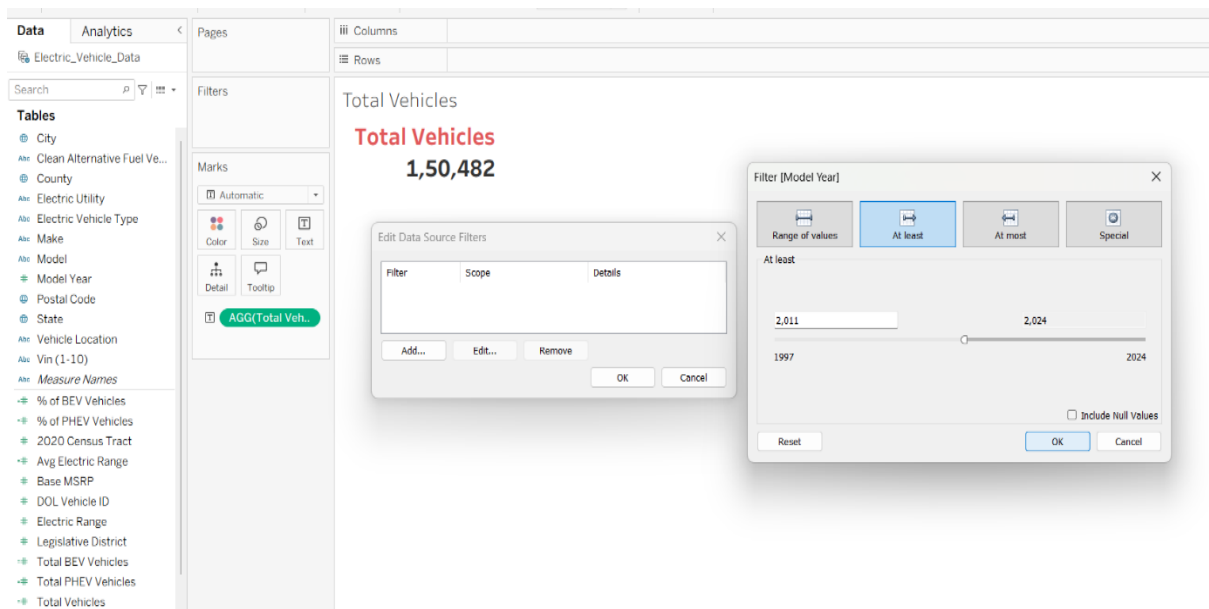


After creating Total Vehicles we need Total Vehicles from 2011 or 2015 whatever the starting year to the year 2024. For counting Electric Vehicles from 2011 onwards:

- Right click on **Electric Vehicle Data Source** on below **Data**.
- Select **Edit Data Source filters**.

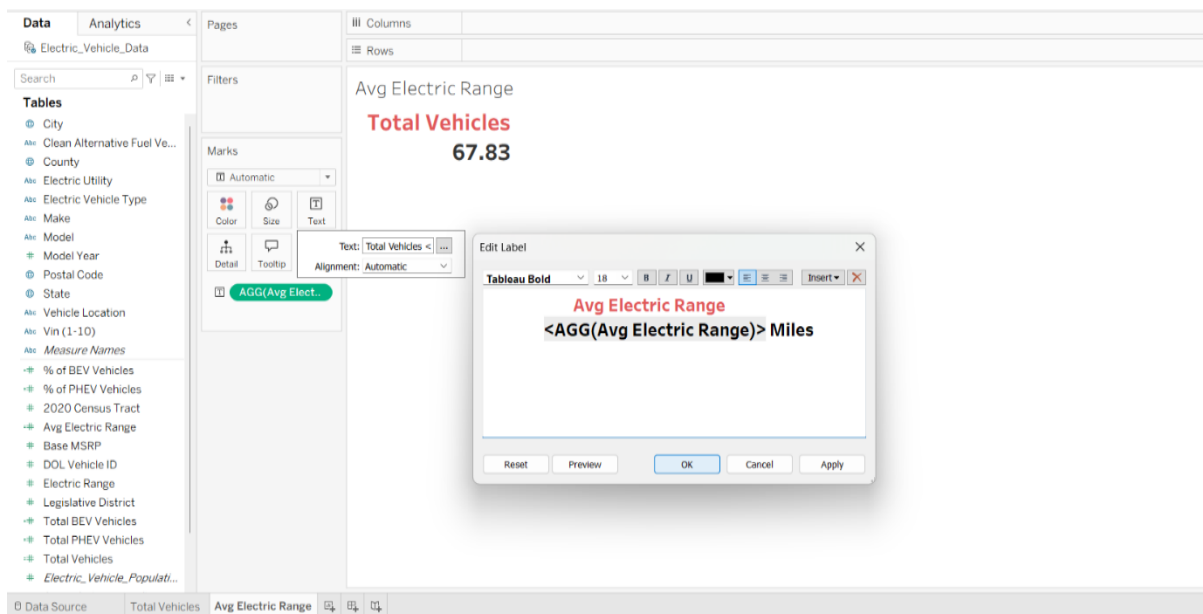


- On **Edit Data Source filter** tab click on **Add** then Select **Model Year** field on Add filter then click **Ok**.
- Model Year Filter tab will be opened, Select **At least** **Box** then set at least year as **2011** and set at most year as **2024** then click on **Ok**.
- Model Year will updated to Filter column on Edit Data Source filter tab, click on **Ok**.
- We can see the small changes in the number of Total vehicles **1,50,482** is decreases to **1,50,422** after setting 2011 is the starting year.



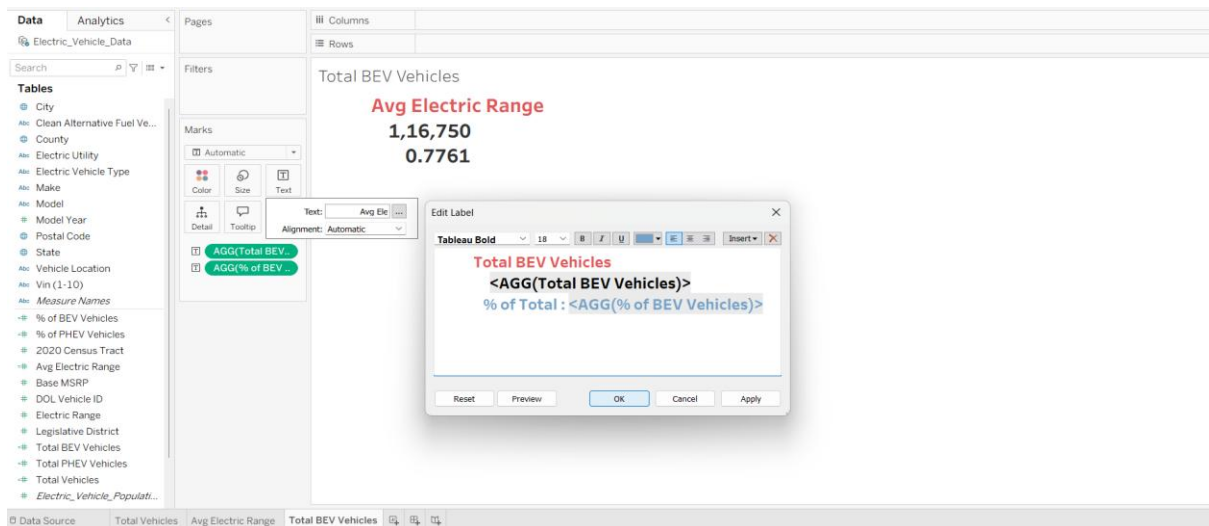
Sheet 2:

- Next we have to create Sheet for **Avg Electric Range** so instead of doing the above process again, we can **duplicate** the Total Vehicles sheet.
- **Right click** on below Total Vehicles sheet name then Select **Duplicate**.
- Rename the duplicate sheet name as **Avg Electric Range**, then drag Avg Electric Range Data field exactly on top of the Total Vehicles data field on Marks bar when black mark will appeared.
- Click on text then click on three dots then change name as Avg Electric range and We want Avg Electric Range in **Miles** so we have to add miles in edit labels, clock on **Ok**.



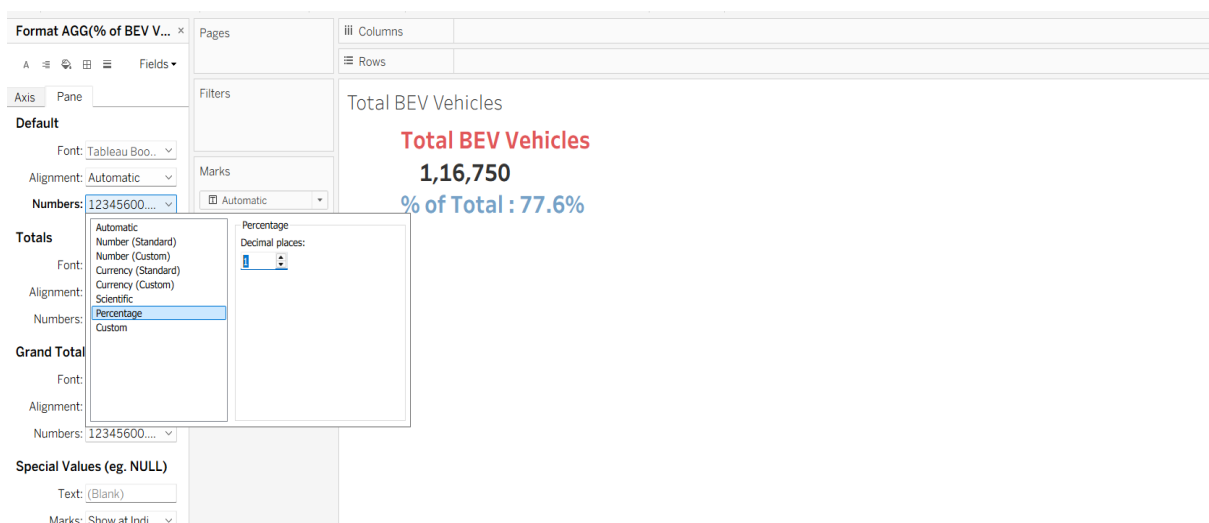
Sheet 3:

- similarly duplicate another sheet and Rename it as **Total BEV Vehicles**.
- Drag Total BEV Vehicles data field exactly on previous data field on Marks. And Drag **% of BEV Vehicles** into Text.
- Go to Edit Label, Change name as **Total BEV Vehicles**. The click on **Ok**.



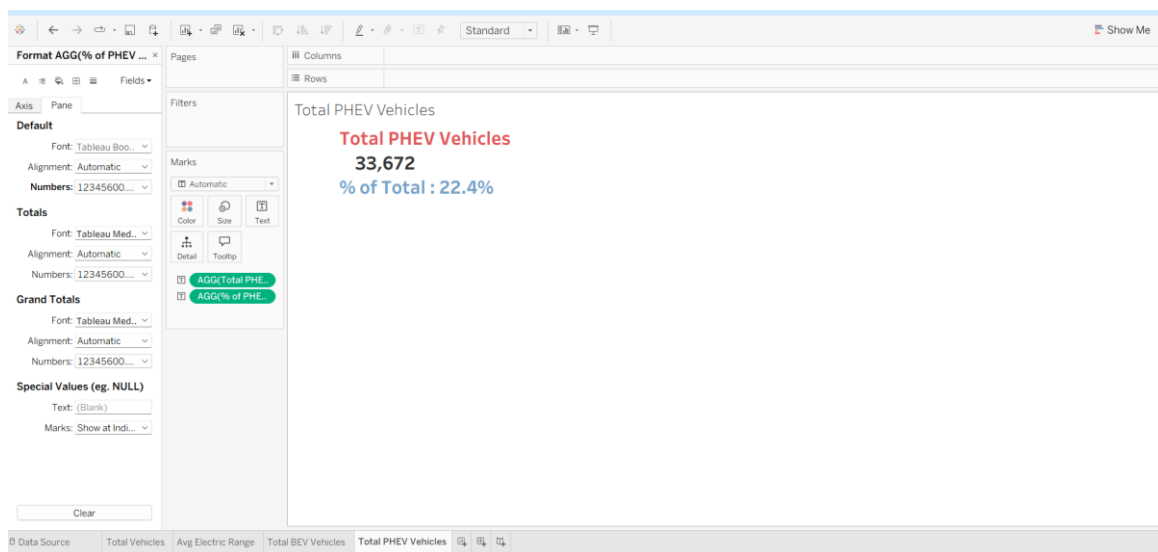
To show BEV Vehicles Total Percentage, **Right click** on **% of BEV Vehicles** Data field Which added into Text on Marks bar.

Select **Format**, Pane Tab will be opened, under **Default** Click on **Numbers (123456)**. Select **Percentage**. Then set Decimal Place into **1**.



Sheet 4:

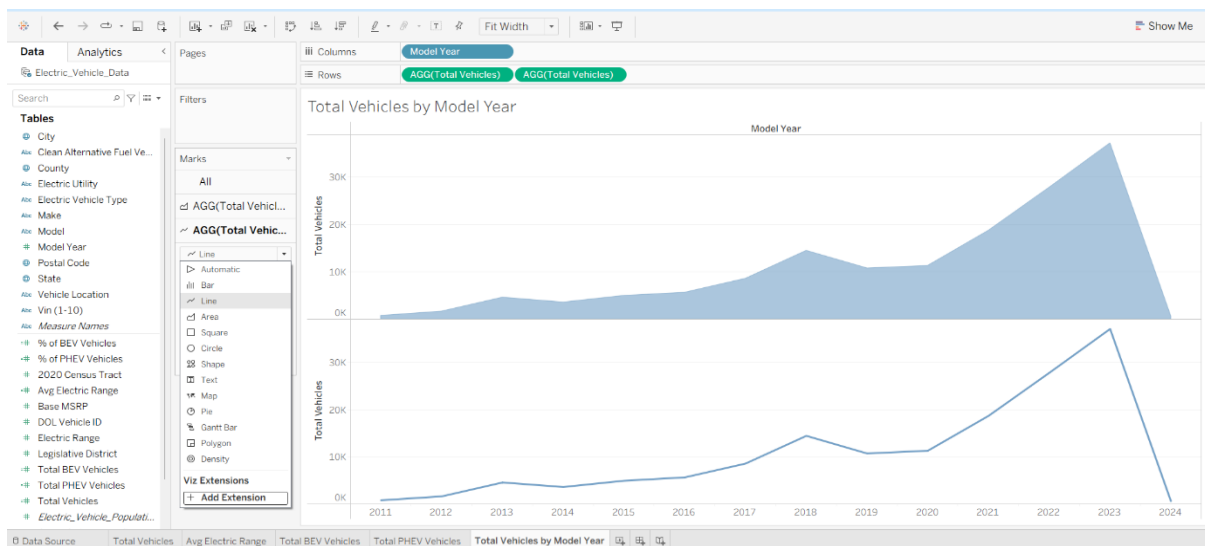
- Duplicate the sheet and Rename it as **Total PHEV Vehicles**. Drag **Total PHEV Vehicles** on BEV Vehicles and **% of PHEV Vehicles** on % of BEV Vehicles.
- Go to Edit Labels rename as **Total PHEV Vehicles**. Then click on **Ok**.
- **Right click** on **% of PHEV Vehicles** Data field Which added into Text. Select **Format** then click on **numbers** then select **Percentage** and set decimal to **1**.



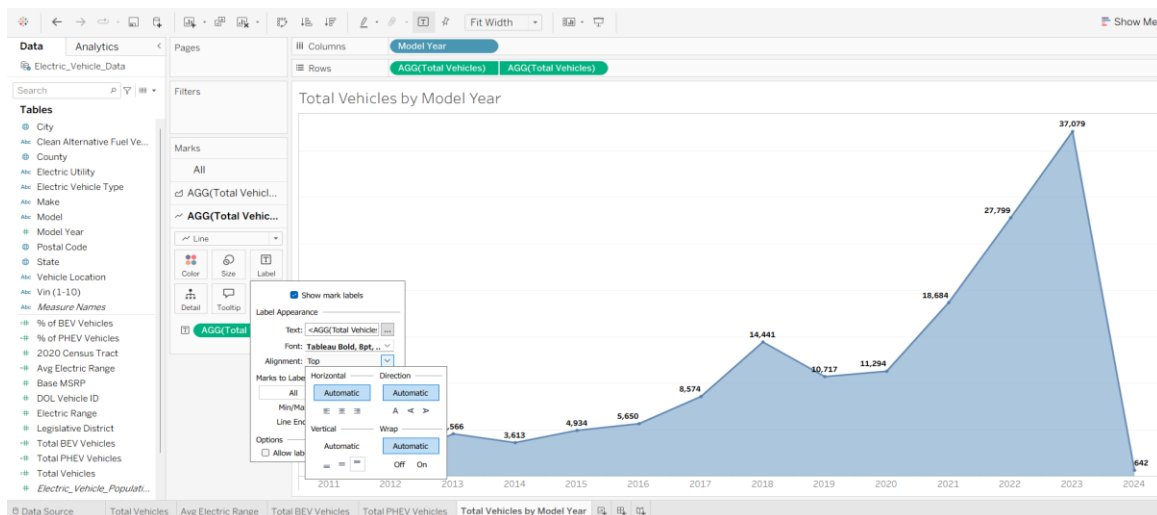
Visualize Electric Vehicles adoption and Trends using Charts for Model Year, States, Manufacturer, Alternative Fuel Vehicle Eligibility, and Consumer Models.

Sheet 5:

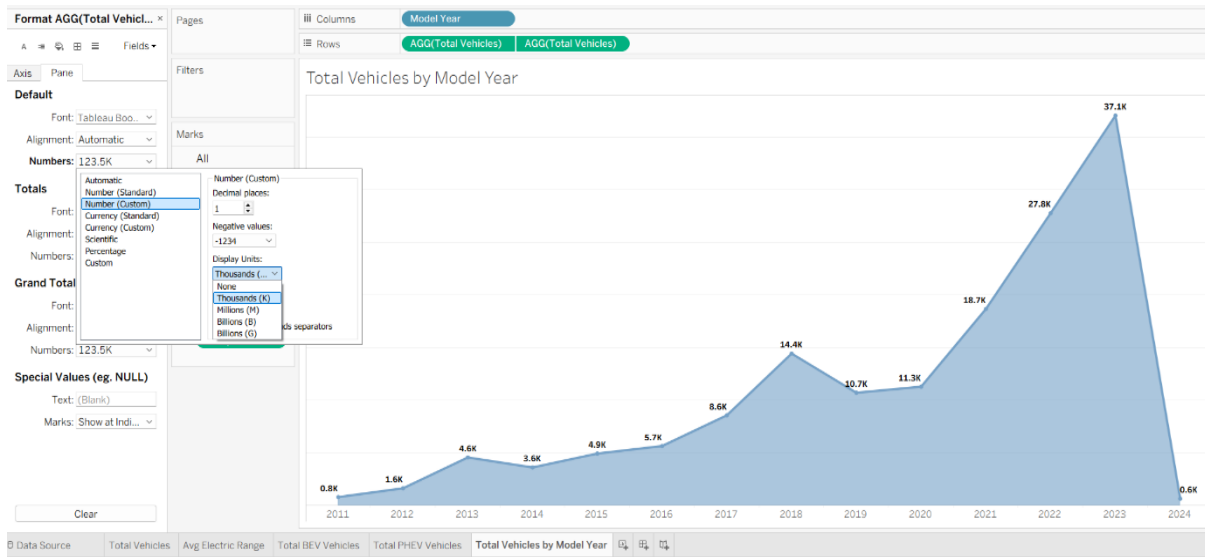
- Create new sheet, Name that as **Total Vehicles by Model Year**.
- Drag **Model Year** to column Shelf and **Total Vehicles** to Row Shelf. The Graph will appears.
- We want Model Years from 2011 to 2024, **Right click** on Model Year which was added into column Shelf, Select **Discrete**. Then click on **Automatic** on Marks bar, Select **Area**.
- Drag Total Vehicles data field again into Row Shelf, Two charts are Created, on Marks bar go to **second Total Vehicles** tab, then click on Area, Select **Line**.



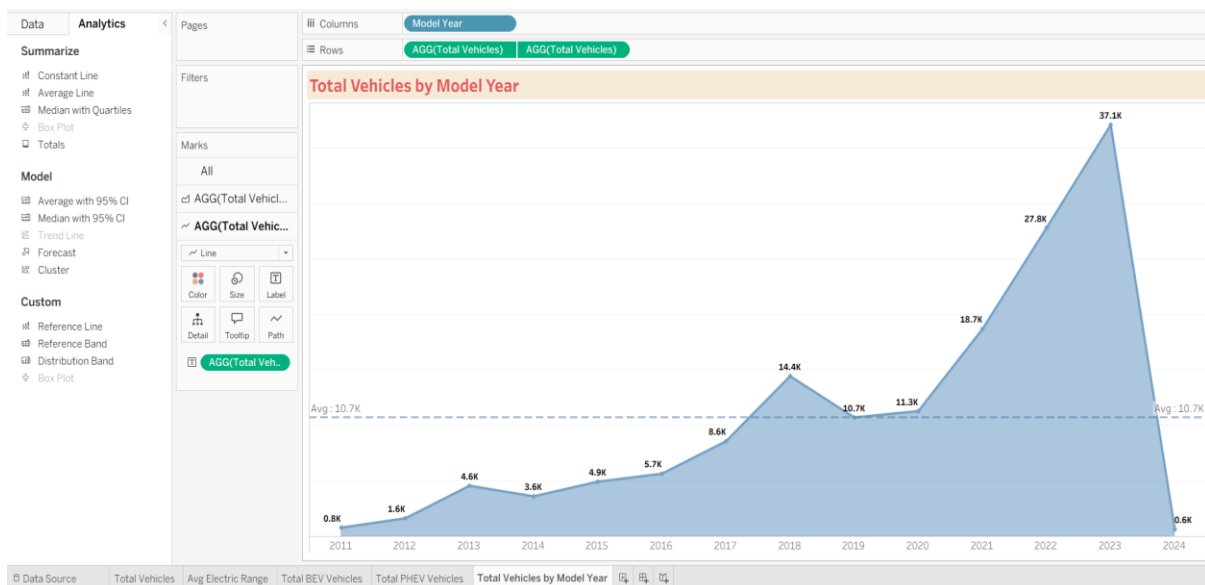
- After **Right click** on second added Total Vehicles field on Row Shelf, Then Select **Dual Axis**. Both charts are merged together.
- Drag Total Vehicles data field into **Labels** on Marks bar. Then click on **Labels**, then click on Alignment, Select **Top** on Vertical.



- **Right click** on Dragged Total Vehicles to label on Marks bar. Then Select **Format**.
- Under Default click on **Numbers (123456)**, then click on **Number (Custom)**, then set Decimal place to **1**, click on **None** under Display units, Select **Thousands (k)**.

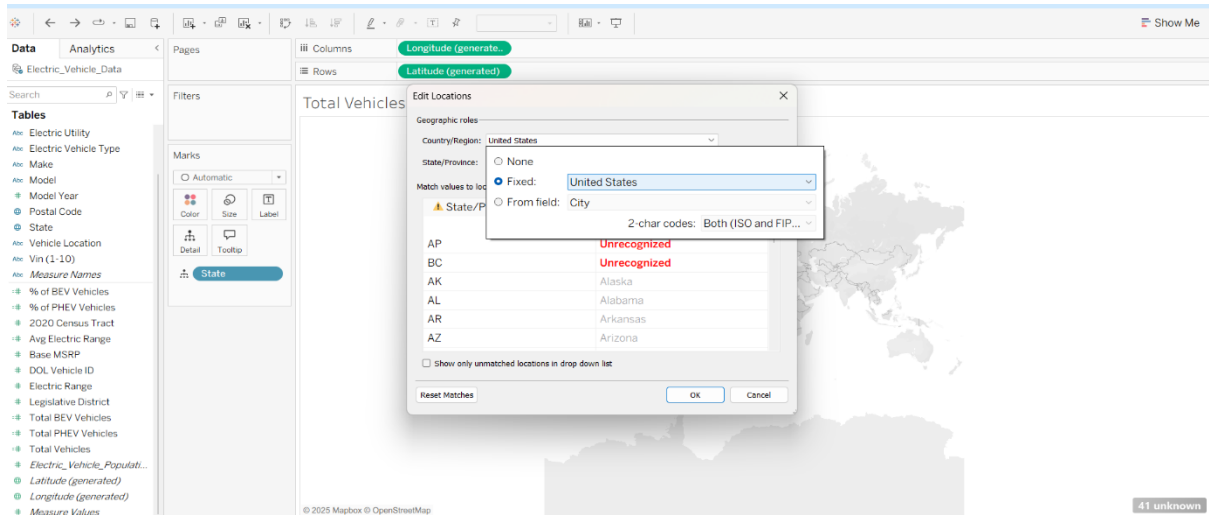


- To make Average Line Click on **Analytics** on Top left side, Drag the **Average Line on Graph** an average line will be created. **Right click** on that created line, then Select **Edit**, edit tab will opened.
- Select **Entire Table** under Scope then click on **Computation** in Label, then Select **Custom**, an empty space will created **type Avg** then click on **greater than (>)** symbol, Select **Value** and Select Line as **Dotted Line (---)** then click on **Ok**.

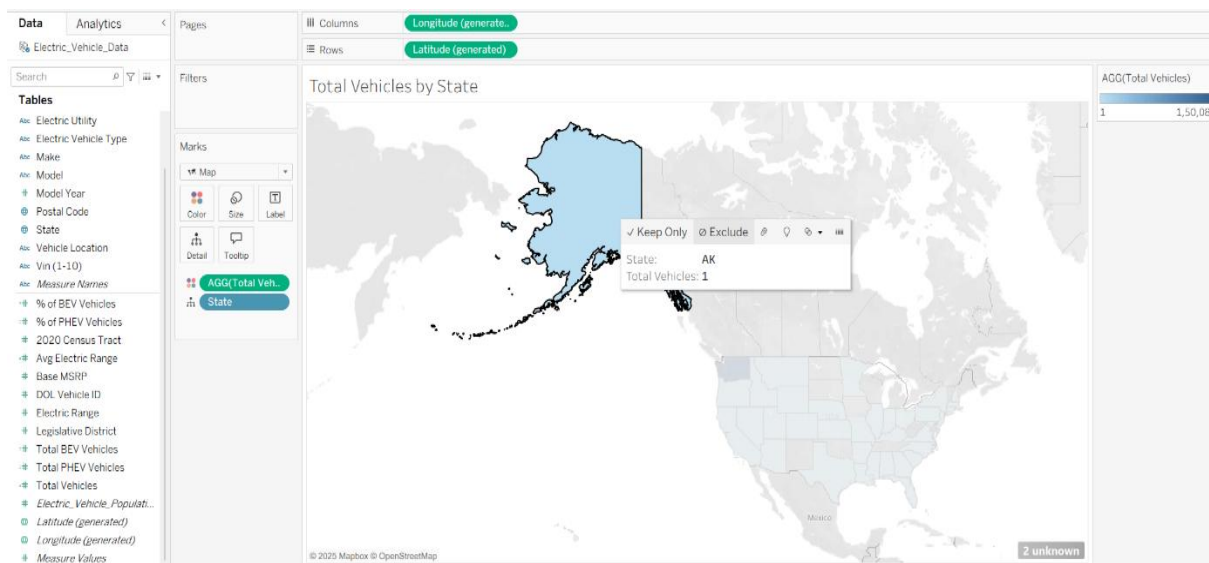


Sheet 6:

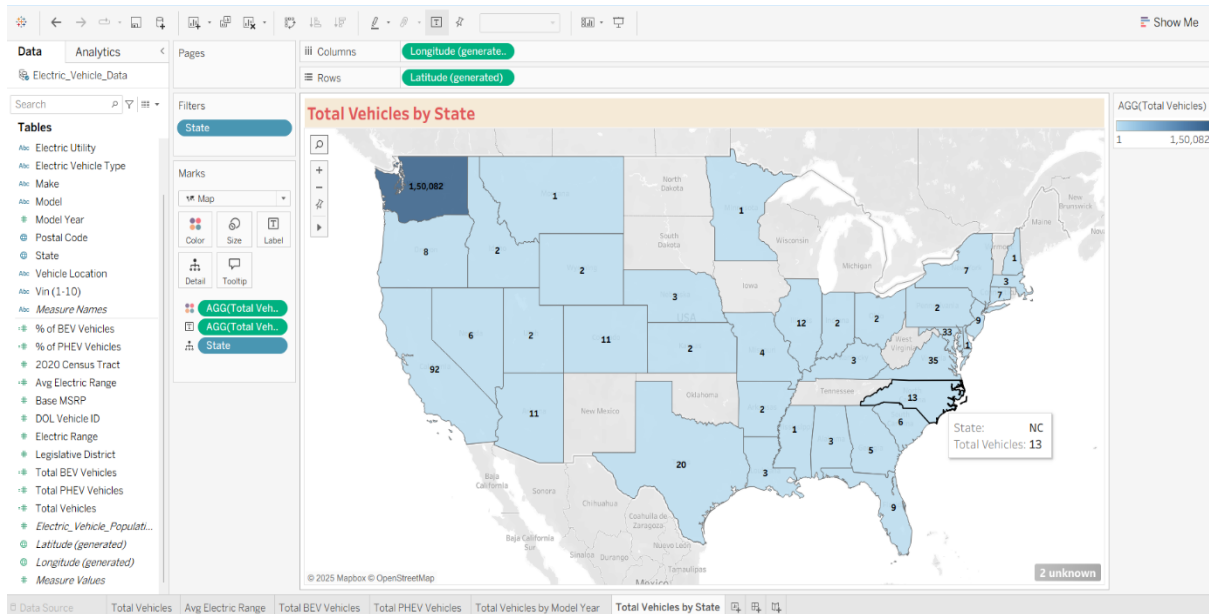
- Create New sheet and name that as **Total Vehicles by State**. Then Drag **Longitude (generated)** into Column Shelf, and **Latitude (generated)** into Row Shelf.
- Drag **State** data field into **Detail** on Marks. Unknown Map will created, **Right click** on the Map then click on **Edit Location**.
- Under country click on **India** then Select **United States**, then click on **Ok**.



On marks click on **Automatic**, select **Map** and drag **Total Vehicles** field into Color on marks. Two unknown States are also created. So click on **AK** and **HI** States then click on **Exclude**.



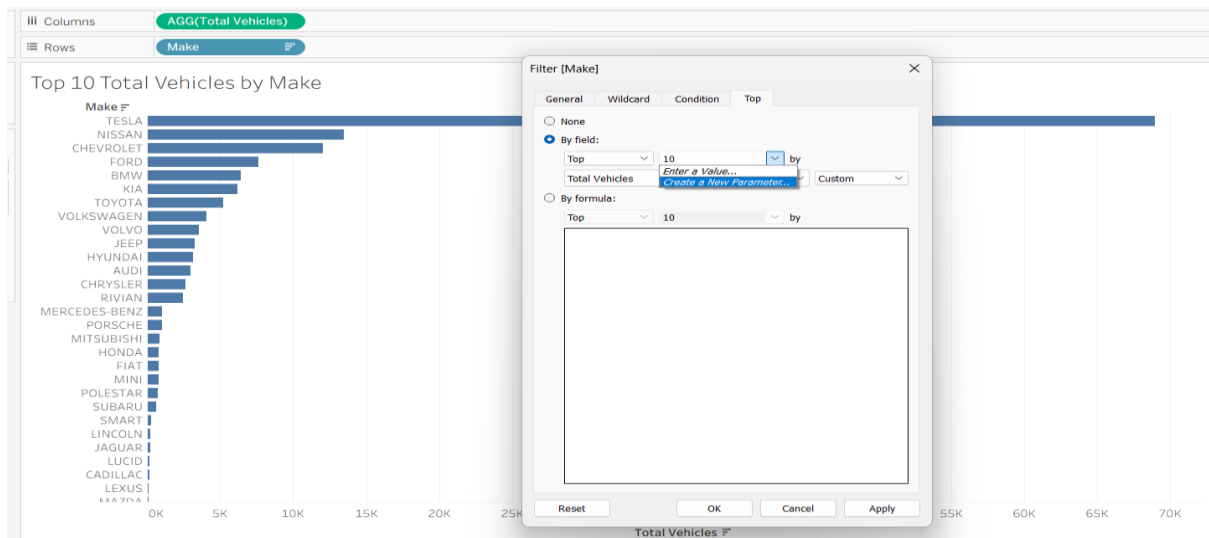
Drag Total Vehicles into **Labels**. Then click on **Label**, click **three dots (...)** under Text. Select Font size and Bold.



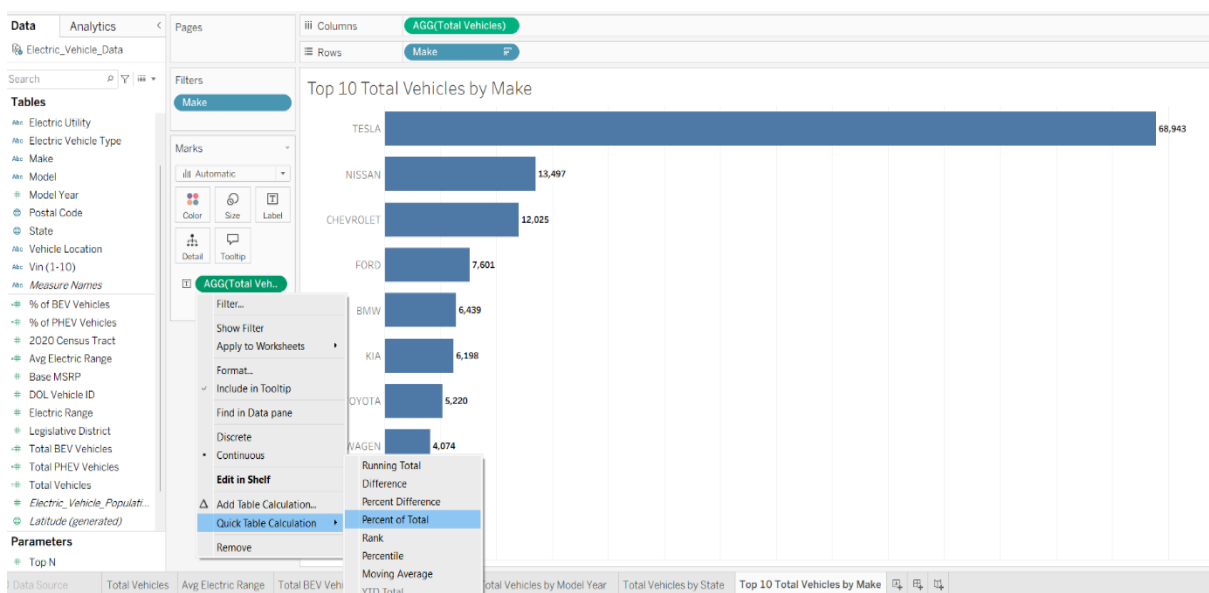
Sheet 7:

Create a New Sheet as **Top 10 Total Vehicles by Make**, Drag **Total Vehicles** into Column and **Make** into Row. Right click on Make on Row Shelf, sort the chart as **Descending Order**.

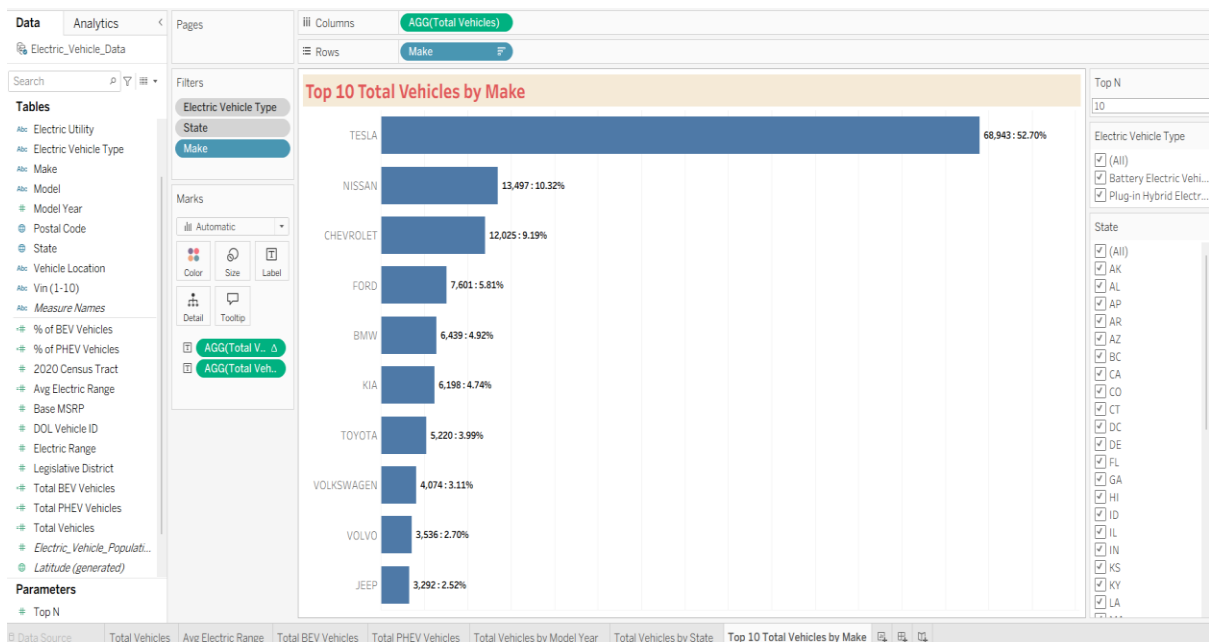
Drag **Make** into **Filter** bar. Filter tab will opened, go to **Top** Condition, then Select **By field**. Click on **10** in front of top, then select **Create a New Parameter**.



- Create Parameter tab will created, Name **Parameter 1** as **Top N**, then Change Maximum **100** as **15** (whatever the Max value). Then click on **Ok**.
- Drag **Total Vehicles** into Label on Marks. Right click on that **Total Vehicles** field, then click on **Quick Table Calculation** and select **Percent of Total**.

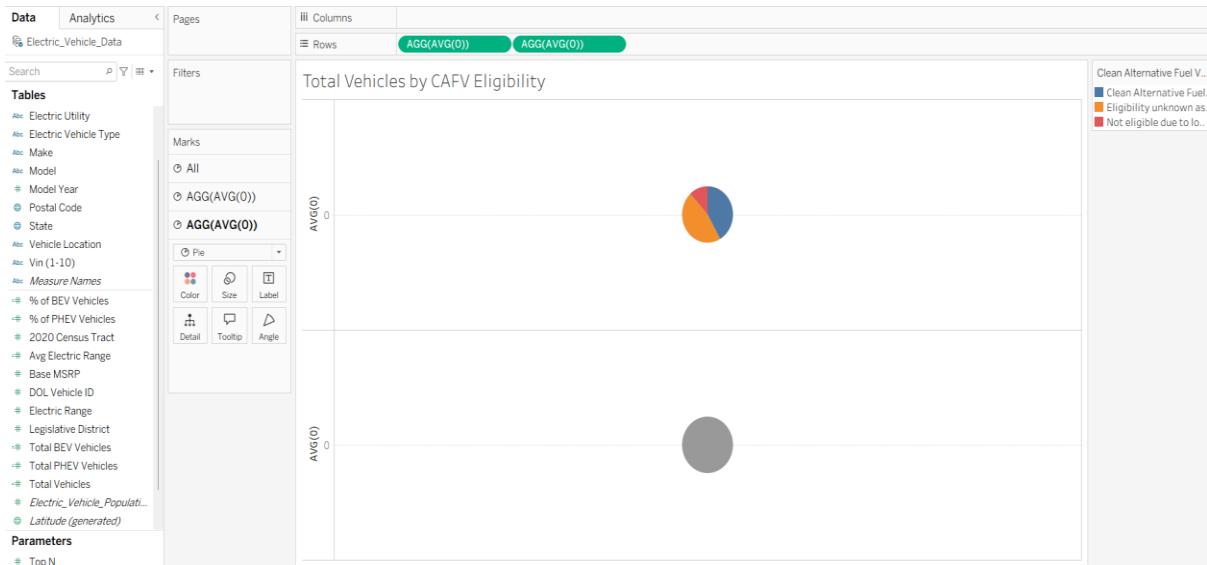


- Again Drag Total Vehicles into Labels. Then click on Label and click on (...), arrange Alignment and Font.
- Right click on **Top N Parameter** left side below, select **Show Parameter**. Top N parameter will appear on right side top corner. **Right click on Top N** then select **Type In.** and type the value from **1 to 15 (whatever we want)**.
- Drag **State** and **Electric Vehicle Type** into Filter. On filter tab select **Use all** for both, Then click on **Ok**. **Right click on State and Electric Vehicle Type** on Filter, select **Show filter**. We can select the State and vehicle Type which we want.
- After we need Vehicle Type and State that what value we give to Top N parameter, so Right click on **State and Electric Vehicle Type on filter** then select **Add to context**.

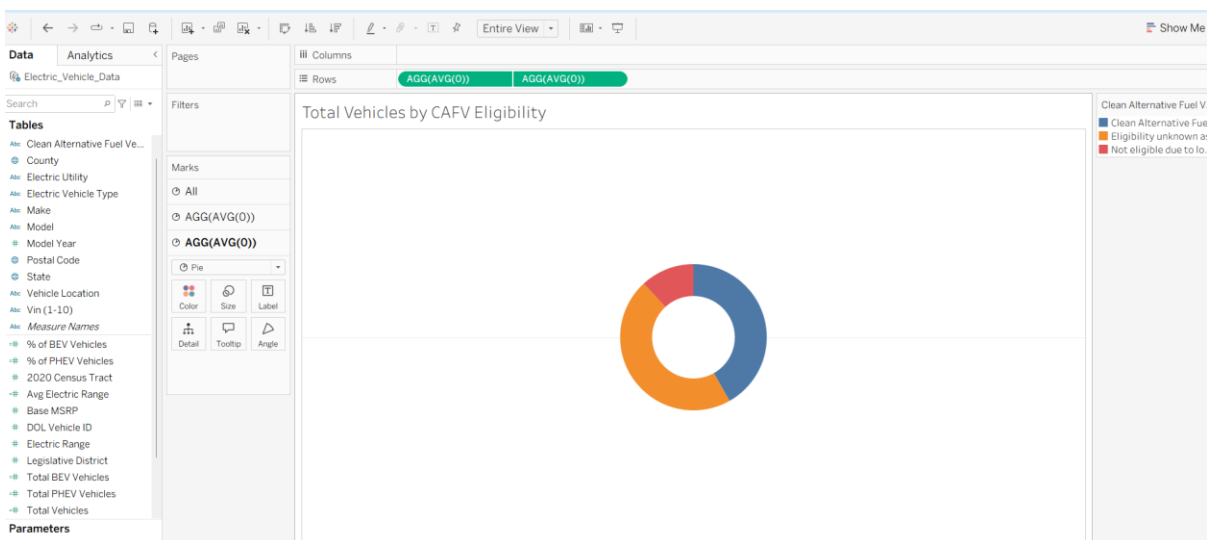


Sheet 8:

- Create a new sheet Name as **Total Vehicles by CAFV Eligibility**. Drag **Clean Alternative Fuel Vehicle (CAFV) Eligible** into Color and click on **Automatic** then select **Pie**. Drag **Total Vehicles** into Angle on Marks.
- Double click on **Row Shelf** and Type **AVG (0)** two times and Enter, Axis will be created. Remove **CAFV** and **Total Vehicles** from second AVG (0) on Marks.



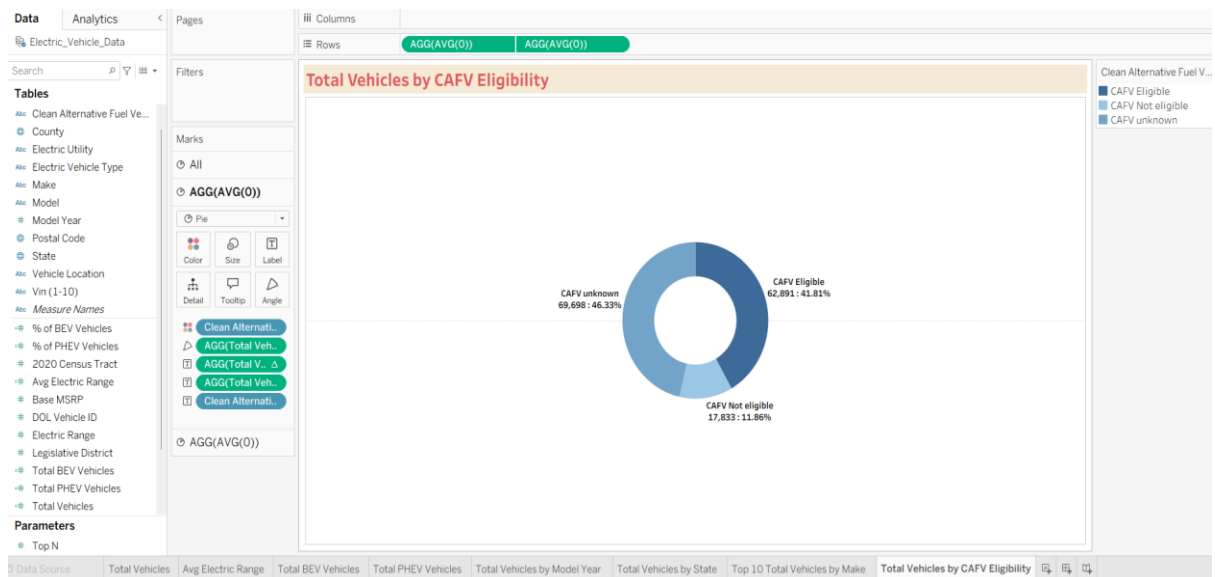
- **Right click** on second **AVG (0)**, select **Dual Axis**. And Increase the **Size** of first AVG (0) on Marks. Click on **Color** on second AVG (0) Marks bar. Select **White**.



Drag **Total Vehicles** into Label on first AVG (0) Marks. Right click on **Total Vehicles**, then click on **Quick Table Calculation** and select **Percent of Total**.

Drag again **Total Vehicles** into Label and drag **CAFV** into Label.

Right click on CAFV data field on Tables. Select **Alias**. And simplify the names. Then click on **Ok**. Click on Label and arrange Alignment. Then Edit colors.

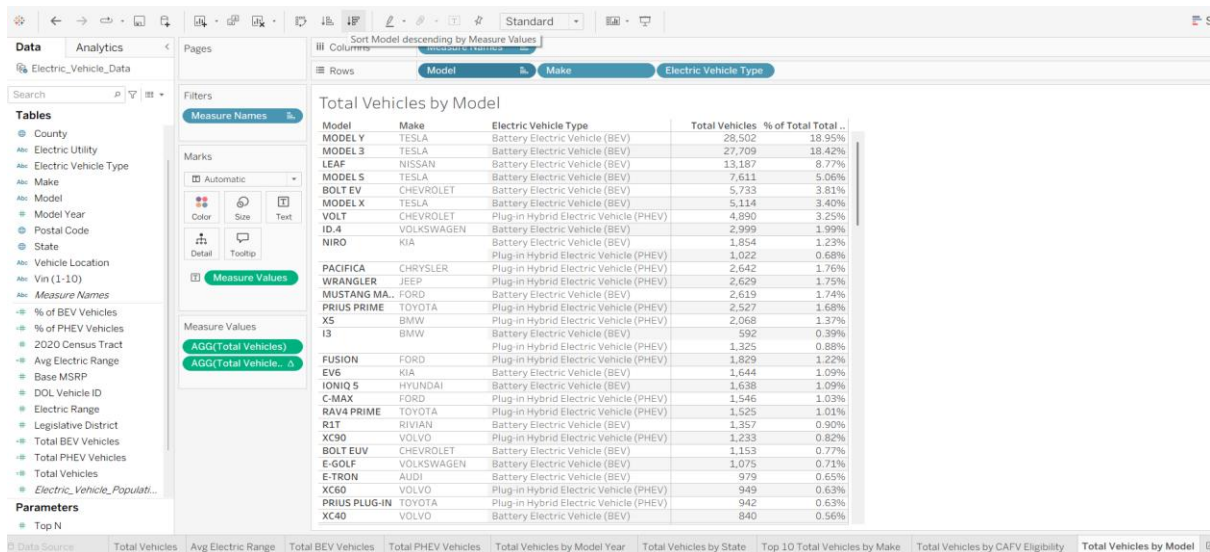


Sheet 9:

Create sheet name as **Total Vehicles by Model**. and Drag **Model, Make and Electric Vehicles Type** into Row Shelf. And Drag **Total Vehicles** into Text on Marks.

We want Total Vehicles in Percentage so **Right click on Total Vehicles** on Marks. Then click on **Quick Table Calculation** then Select **Percent of Total**. Double click on **Total Vehicles** data field on Tables bar.

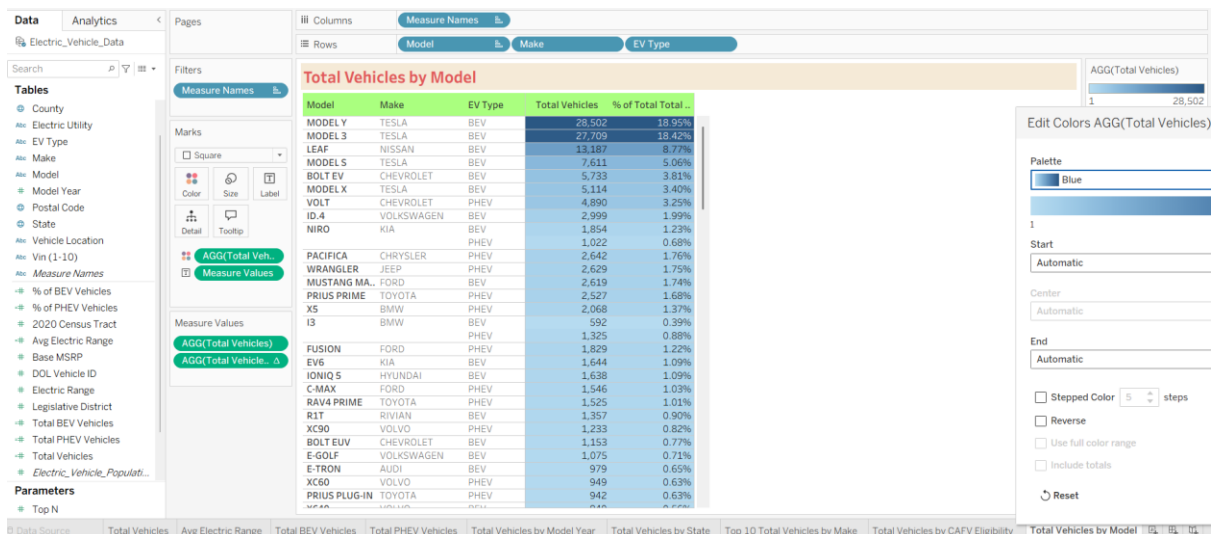
Just Click on **Model** on and Sort in **Descending Order**.



Model	Make	Electric Vehicle Type	Total Vehicles	% of Total Total
MODEL Y	TESLA	Battery Electric Vehicle (BEV)	28,502	18.95%
MODEL 3	TESLA	Battery Electric Vehicle (BEV)	27,709	18.42%
LEAF	NISSAN	Battery Electric Vehicle (BEV)	13,187	8.77%
MODEL S	TESLA	Battery Electric Vehicle (BEV)	7,611	5.06%
BOLT EV	CHEVROLET	Battery Electric Vehicle (BEV)	5,733	3.81%
MODEL X	TESLA	Battery Electric Vehicle (BEV)	5,114	3.40%
VOLT	CHEVROLET	Plug-in Hybrid Electric Vehicle (PHEV)	4,890	3.25%
ID.4	VOLKSWAGEN	Battery Electric Vehicle (BEV)	2,999	1.99%
NIRO	KIA	Plug-in Hybrid Electric Vehicle (PHEV)	1,854	1.23%
			1,022	0.68%
PACIFICA	CHRYSLER	Plug-in Hybrid Electric Vehicle (PHEV)	2,642	1.76%
WRANGLER	JEEP	Plug-in Hybrid Electric Vehicle (PHEV)	2,629	1.75%
MUSTANG MA.	FORD	Battery Electric Vehicle (BEV)	2,619	1.74%
PRIUS PRIME	TOYOTA	Plug-in Hybrid Electric Vehicle (PHEV)	2,527	1.68%
XS	BMW	Plug-in Hybrid Electric Vehicle (PHEV)	2,068	1.37%
I3	BMW	Battery Electric Vehicle (BEV)	592	0.39%
			1,325	0.88%
FUSION	FORD	Plug-in Hybrid Electric Vehicle (PHEV)	1,829	1.22%
EVE	KIA	Battery Electric Vehicle (BEV)	1,644	1.09%
IONIQ 5	HYUNDAI	Battery Electric Vehicle (BEV)	1,638	1.09%
C-MAX	FORD	Plug-in Hybrid Electric Vehicle (PHEV)	1,546	1.03%
RAV4 PRIME	TOYOTA	Plug-in Hybrid Electric Vehicle (PHEV)	1,525	1.01%
R1T	RIVIAN	Battery Electric Vehicle (BEV)	1,357	0.90%
XC90	VOLVO	Plug-in Hybrid Electric Vehicle (PHEV)	1,233	0.82%
BOLT EUV	CHEVROLET	Battery Electric Vehicle (BEV)	1,153	0.77%
E-GOLF	VOLKSWAGEN	Battery Electric Vehicle (BEV)	1,075	0.71%
E-TRON	AUDI	Battery Electric Vehicle (BEV)	979	0.65%
XC60	VOLVO	Plug-in Hybrid Electric Vehicle (PHEV)	949	0.63%
PRIUS PLUG-IN	TOYOTA	Plug-in Hybrid Electric Vehicle (PHEV)	942	0.63%
XC40	VOLVO	Battery Electric Vehicle (BEV)	840	0.56%

we want to Simplify the **Electric Vehicle Type**, so Right Click on **Electric Vehicle Type** and click on **Alias** then simplify the names **BEV** and **PHEV**.

Drag **Total Vehicles** into color then click on **Automatic** and select **square**. then edit colours. Right click on **Model** column Header on Table and select **Format** then select **Shadings**.



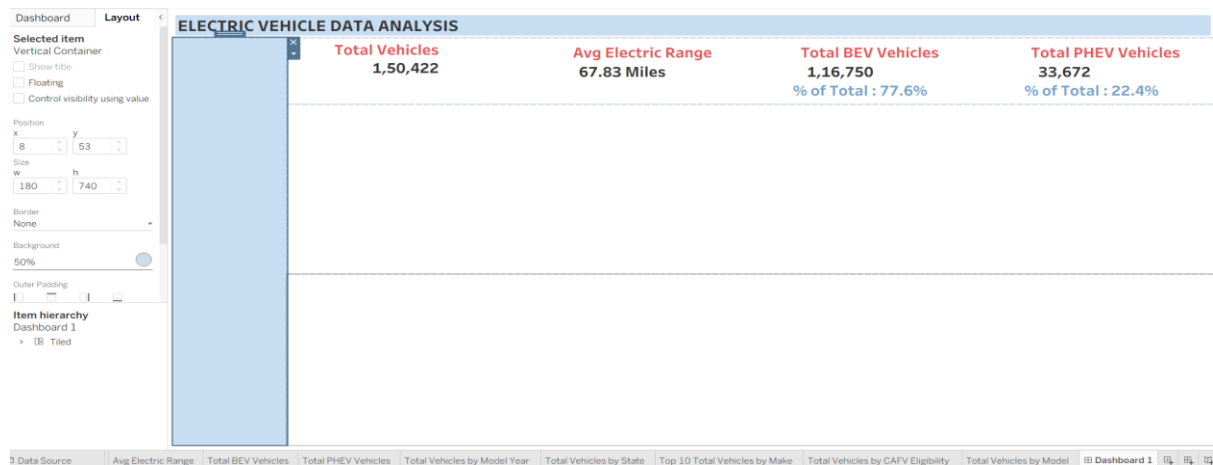
Model	Make	EV Type	Total Vehicles	% of Total Total
MODEL Y	TESLA	BEV	28,502	18.95%
MODEL 3	TESLA	BEV	27,709	18.42%
LEAF	NISSAN	BEV	13,187	8.77%
MODEL S	TESLA	BEV	7,611	5.06%
BOLT EV	CHEVROLET	BEV	5,733	3.81%
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PRIUS PLUG-IN	TOYOTA	PHEV	942	0.63%
XC40	VOLVO	BEV	840	0.56%

Step 4 : Create Dashboard

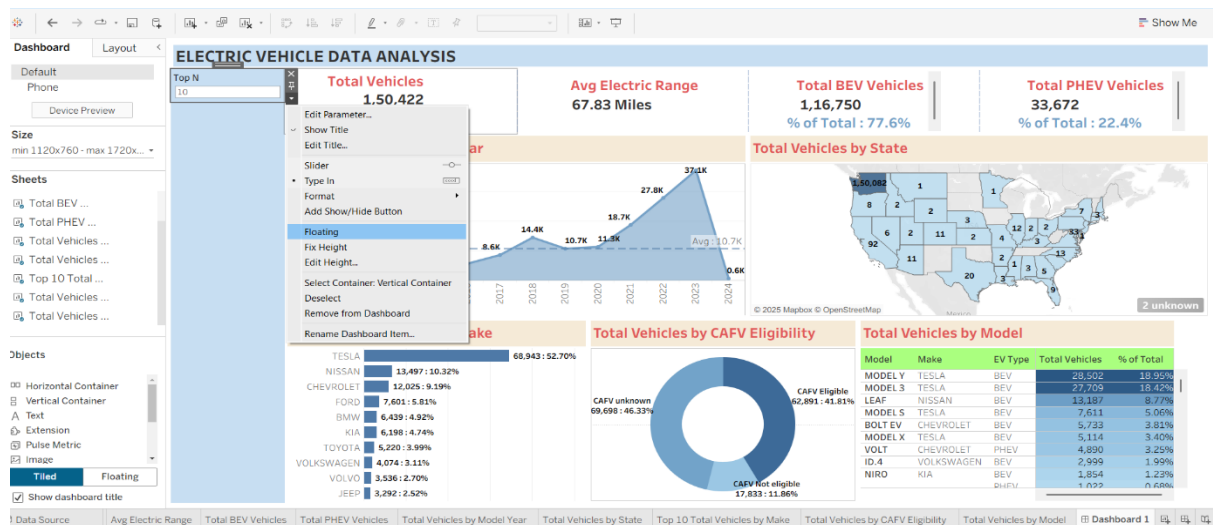
Dashboard name as **Electric Vehicle Data Analysis**. Drag one **Horizontal container** and one **Blank** from Objects.

Drag starting 4 Sheets into **Horizontal container** then click on **More option** select **Distribute Content Evenly**.

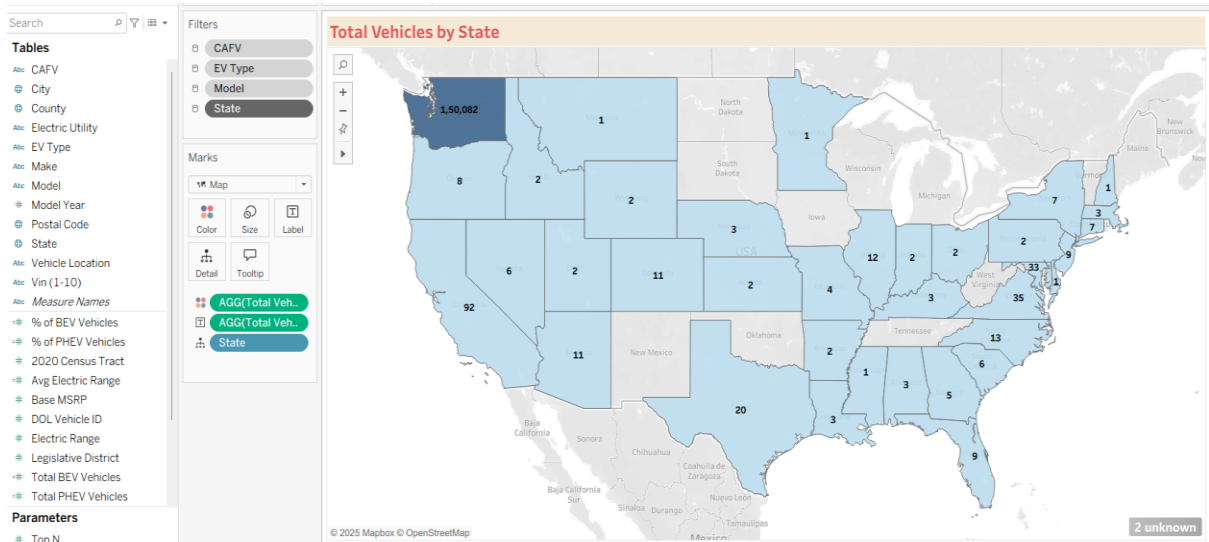
Drag **Vertical container** into left side of dashboard. Then click on **Layout** and select the **Background colour**.



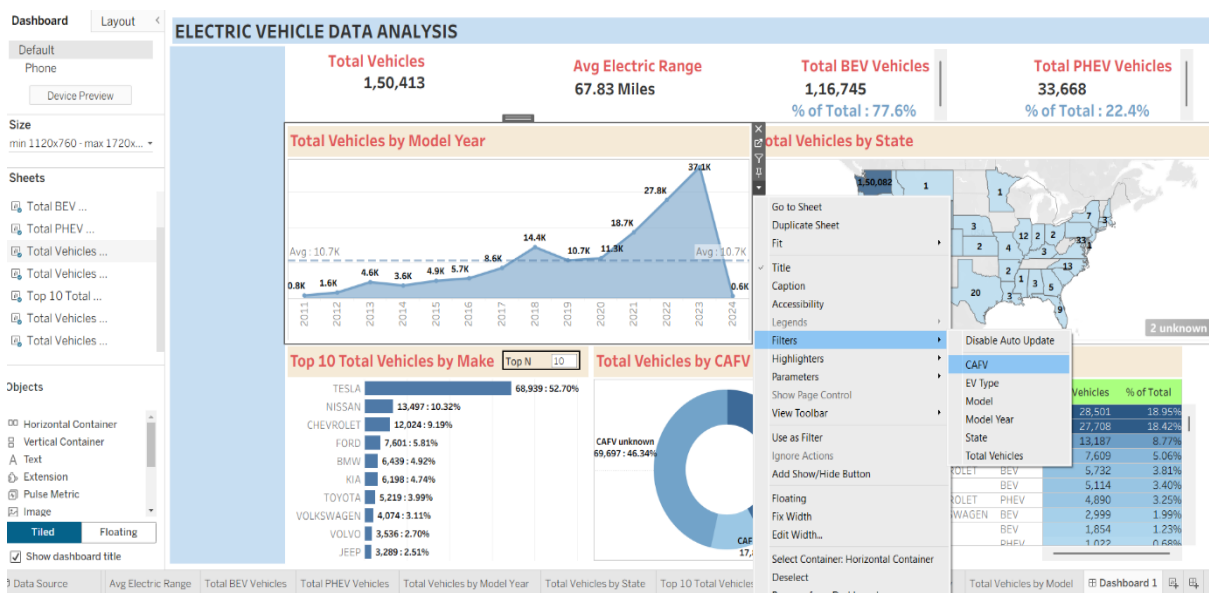
- Drag Sheets that Total Vehicles by Model Year, State, Make, CAFV eligibility and Model.
- Remove all the filters Except **Top N**, click on More option on TOP N filter and select **Floating**. Then drag that Top N filter to **Top 10 Total Vehicles by Make**.
- Again select more option click on **Format** and make Border.



- Go back to the **Total Vehicles by State** Sheet. Drag **CAFV**, **EV Type**, **Model** data fields into filter. And select **Use all** on filter tab then click on **Ok**.
- Right click on **CAFV**, **EV Type**, **Model** and **State**. Select **Add to Context** and Click on **Apply to Worksheet** and select **All Using This Data Source** from all 4 data fields on filter. One icon mark will appear.



Click on **more option** in any charts on Dashboard and click on **Filters** then Select all **CAFV**, **EV Type**, **Model**, and **State** filters.



Click on **More options** on Added all filters, select **Multiple Values (Dropdown)** for all.

Click on **Top 10 Total Vehicles by Make** chart and select **Use as Filter** option above the More option.

